

5.1. ASSET ALLOCATION WITH TRANSACTIONS COSTS

George Constantinides (1986) was the first to develop an asset allocation model where the investor had to pay transaction costs. Selling \$100 of equities, for example, results in a final position of \$90 with 10% transactions costs. Not surprisingly, the investor trades infrequently—to save on transactions costs. Constantinides proved that the optimal strategy is to trade whenever risky asset positions hit upper or lower bounds. Within these bounds is an interval of no trading. The no-trading band straddles the optimal asset allocation from a model that assumes you can continuously trade without frictions (the Merton 1971 model; see chapter 4).⁴²

The no-trade interval is a function of the size of the transactions costs and the volatility of the risky asset. Constantinides estimates that for transactions costs of 10%, there are no-trade intervals greater than 25% around an optimal holding of 25% for a risky asset with a 35% volatility. (I bet Harvard wished it could have received just a 10% discount when it tried to sell its private equity investments in 2008.) That is, the asset owner would not trade between (0%, 50%)—indeed, very large fluctuations in the illiquid asset position. Illiquid asset investors should expect to rebalance very infrequently.

Constantinides' model can be used to compute an illiquidity risk premium, defined as the expected return of an illiquid asset required to bring the investor to the same level of utility as in a frictionless setting. This is the risk premium the investor demands to bear the transactions costs and is a certainty equivalent calculation (see chapter 2). For transaction costs of 15% or more, the required risk premium exceeds 5%. Compare this value with (the close to) zero additional excess returns, on average, of the illiquid asset classes in data.

A major shortcoming of the transaction costs models is that they assume trade is always possible by paying a cost. This is not true for private equity, real estate, timber, or infrastructure. Over a short horizon, there may be no opportunity to find a buyer. Even if a counterparty can be found, you need to wait for due diligence and legal transfer to be completed and then the counterparty can get cold feet.⁴³ Many liquid assets also experienced liquidity freezes during the financial crisis where no trading—at any price—was possible because no buyers could be found.

⁴² Chapter 4 discusses extensions of Constantinides (1986) to double bands, contingent bands, and rebalancing to the edge or center of the bands.

⁴³ For some illiquid assets, investors may not be even willing to transact immediately for one cent; some investments do not have liability limited at zero. For example, on June 30, 2008, a real estate investment by CalPERS was valued at negative \$300 million! See Corkery, M., C. Karmin, R. L. Rundle, and J. S. Lublin, "Risky, Ill-Timed Land Deals Hit CalPERS," *Wall Street Journal*, Dec. 17, 2008.